

2. Prepare a CI/CD workflow

Prepare a CI/CD workflow report describing the pipeline architecture, stages, automation results, and observations with supporting screenshots.

1. **Check Git Installation:** `git --version`
2. **Configure Git Username:** `git config --global user.name "Your Name"`
3. **Configure Git Email:** `git config --global user.email "your-email@example.com"`
4. **Check Git Configuration:** `git config --global --list`
5. **Initialize Git Repository:** `git init`
6. **Check Git Status:** `git status`
7. **Add Project Files:** `git add .`
8. **Create Git Commit:** `git commit -m "Initial Docker project"`
9. **Create Observations File:** `nano observations.txt`
10. **View Commit History:** `git log --oneline`
11. **Add GitHub Remote:** `git remote add origin YOUR_GITHUB_REPOSITORY_URL`
12. **Check GitHub Remote:** `git remote -v`
13. **Rename Branch to Main:** `git branch -M main`
14. **Pull Remote Changes:** `git pull origin main --rebase`
15. **Push Project to GitHub:** `git push -u origin main`
16. **Create GitHub Actions Directory:** `mkdir -p .github/workflows`
17. **Create CI/CD Workflow File:** `nano .github/workflows/docker-ci.yml`
18. **Check Workflow File:** `ls .github/workflows`
19. **Display Workflow Configuration:** `cat .github/workflows/docker-ci.yml`
20. **Add CI/CD Workflow:** `git add .`
21. **Commit CI/CD Workflow:** `git commit -m "Added Docker CI/CD workflow"`
22. **Push CI/CD Workflow:** `git push`
23. **Test Application:** `curl http://localhost:5000`

24. View Git Commit History: git log --oneline

```
ayushi2323@LAPTOP-6LVIDNHT: ~/docker-pr | X + v - □ X
ayushi2323@LAPTOP-6LVIDNHT:~$ cd ~/docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile  commands.txt  main.py  observations.txt  requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
nothing to commit, working tree clean
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git remote add origin https://github.com/ayushi2323/Devops-lab.git
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git
Username for 'https://github.com': ayushi2323
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ [PASSWORD-INPUT-PLACEHOLDER]
Password for 'https://ayushi2323@github.com':
DER]
Enumerating objects: 5, done.
Counting objects: 100% (15/5), done.
Username for 'https://github.com': ayushi2323
Password for 'https://ayushi2323@github.com':
Compressing objects: 100% (15/5), done.
Writing objects: 100% (15/5), 450 bytes | 4500 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ayushi2323/Devops-lab.git
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': ayushi2323
Password for 'https://ayushi2323@github.com': [PASSWORD-INPUT-PLACEHOLDER]
DER]
branch 'main' set up to track 'origin/main'.
Enumerating objects: 5, done.
Delta compression using up to 8 threads
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 450 bytes | 450.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ayushi2323/Devops-lab.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ |
```

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git pull origin main --rebase
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (10/10), 4.38 MiB | 3.01 MiB/s, done.
From https://github.com/ayushi2323/Devops-lab
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Successfully rebased and updated refs/heads/main.
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': ayushi2323-CICD
Password for 'https://ayushi2323-CICD@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (6/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.31 KiB | 1.31 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayushi2323/Devops-lab
   lacaal6, db2aaf6  main -> main
branch 'main' set up to track 'origin/main'.
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ mkdir -p .github/workflows
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls -la
total 4868
drwxr-xr-x  4 ayushi2323 ayushi2323  4096 Sep  9 13:49 .
drwxr-xr-x 13 ayushi2323 ayushi2323  4096 Sep  9 13:12 ..
drwxr-xr-x  7 ayushi2323 ayushi2323  4096 Sep  9 13:49 .git
drwxr-xr-x  3 ayushi2323 ayushi2323  4096 Sep  9 13:49 .github
-rw-r--r--  1 ayushi2323 ayushi2323   164 Sep  9 13:48 Dockerfile
-rw-r--r--  1 ayushi2323 ayushi2323 3669414 Sep  9 13:48 'LAB2 TASK2.pdf'
-rw-r--r--  1 ayushi2323 ayushi2323  743149 Sep  9 13:48 'Lab2 (1).docx'
-rw-r--r--  1 ayushi2323 ayushi2323    12 Sep  9 13:48 README.md
-rw-r--r--  1 ayushi2323 ayushi2323   301 Sep  9 13:48 commands.txt
-rw-r--r--  1 ayushi2323 ayushi2323 526460 Sep  9 13:48 lab2.c
-rw-r--r--  1 ayushi2323 ayushi2323   197 Sep  9 13:48 main.py
-rw-r--r--  1 ayushi2323 ayushi2323   762 Sep  9 13:48 observations.txt
```

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git pull origin main --rebase
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (10/10), 4.38 MiB | 3.01 MiB/s, done.
From https://github.com/ayushi2323/Devops-lab
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Successfully rebased and updated refs/heads/main.
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Ayushi-CICD
Password for 'https://Ayushi-CICD@github.com': [PASSWORD-HIDDEN]
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.31 KiB | 1.31 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayushi2323/Devops-lab
   lacaal6..db2aaf6  main -> main
branch 'main' set up to track 'origin/main'.
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ mkdir -p .github/workflows
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls -la
total 4868
drwxr-xr-x  4 ayushi2323 ayushi2323    4096 Sep  9 13:49 .
drwxr-xr-x 13 ayushi2323 ayushi2323    4096 Sep  9 13:12 ..
drwxr-xr-x  7 ayushi2323 ayushi2323    4096 Sep  9 13:49 .git
drwxr-xr-x  3 ayushi2323 ayushi2323    4096 Sep  9 13:49 .github
-rw-r----- 1 ayushi2323 ayushi2323     164 Sep  9 13:48 Dockerfile
-rw-r----- 1 ayushi2323 ayushi2323 3669414 Sep  9 13:48 'AYUSHI LAB TASK2.pdf'
-rw-r----- 1 ayushi2323 ayushi2323   743149 Sep  9 13:48 'Ayushi Lab2 (1).docx'
-rw-r----- 1 ayushi2323 ayushi2323      12 Sep  9 13:48 README.md
-rw-r----- 1 ayushi2323 ayushi2323     301 Sep  9 13:48 commands.txt
-rw-r----- 1 ayushi2323 ayushi2323 526460 Sep  9 13:48 'ayushi2323-lab2'
-rw-r----- 1 ayushi2323 ayushi2323     197 Sep  9 13:48 main.js
-rw-r----- 1 ayushi2323 ayushi2323     762 Sep  9 13:48 'ayushi2323-observations.txt'
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$
```

```
ayushi2323@LAPTOP-6LVIDN x + v
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ ls .github/workflows
docker-ci.yml
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ cat .github/workflows/docker-ci.yml
name: Docker CI/CD

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app .

      - name: Run Docker container
        run: |
          docker run -d -p 5000:5000 --name my-container my-docker-app
          sleep 5

      - name: Test application
        run: curl http://localhost:5000

      - name: Stop container
        run: docker stop my-container
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
.github/
```

```
ayushi2323@LAPTOP-6LVIDNHT x + v
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls .github/workflows
docker-ci.yml
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ cat .github/workflows/docker-ci.yml
name: Docker CI/CD

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app:v1.0.0 .

      - name: Run Docker container
        run: |
          docker run -d -p 5000:5000 --name my-container my-docker-app:
v1.0.0      sleep 5

      - name: Test application
        run: curl http://localhost:5000

      - name: Stop container
        run: docker stop my-container
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
.github/
```

```
ayushi2323@LAPTOP-6LVIDNHT:~$ ls .github/workflows
ayushi2323_ci.yml
ayushi2323@LAPTOP-6LVIDNHT:~$ cat .github/workflows/ayushi2323_ci.yml

name: Docker CI/CD

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app .

      - name: Run Docker container
        run: |
          docker run -d -p 5000:5000 --name ayushi2323-container my-do
          cker-app sleep 5

      - name: Test application
        run: curl http://localhost:5000

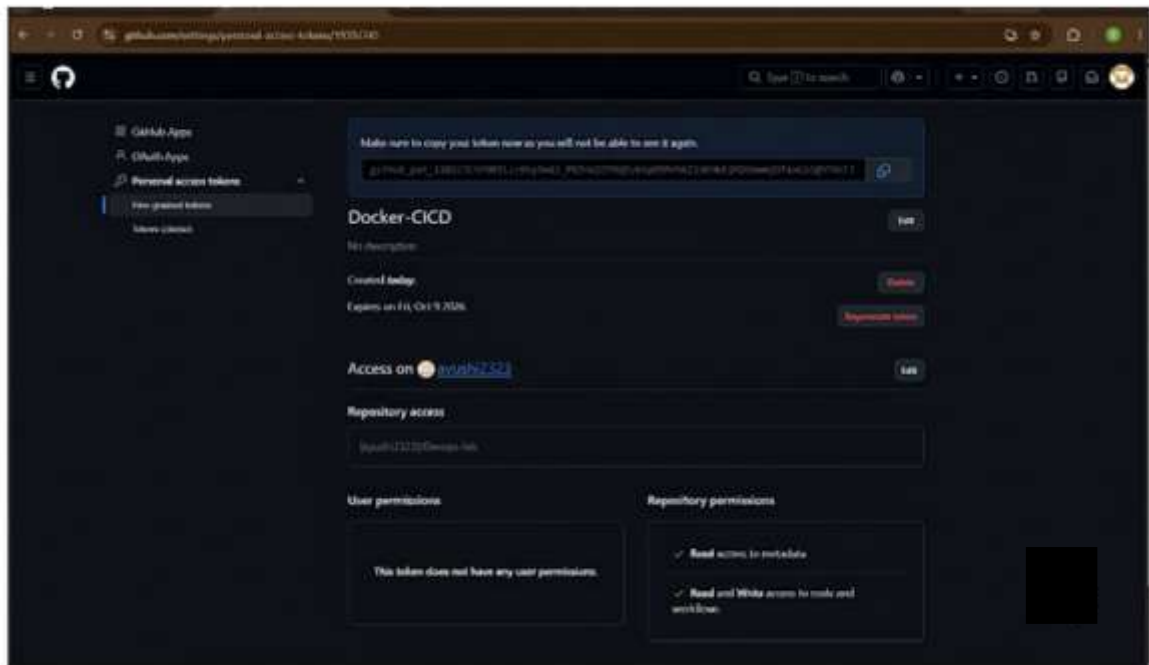
      - name: Stop container
        run: docker stop ayushi2323-container
ayushi2323@LAPTOP-6LVIDNHT:~$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .github/
```

```
ayushi2323@LAPTOP-6LVIDNHT x + v - □ ×

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .github/

nothing added to commit but untracked files present (use "git add" to track)
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git add .
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Added Docker CI/CD workflow"
[main 9e7f00c] Added Docker CI/CD workflow
 1 file changed, 31 insertions(+)
 create mode 106644 .github/workflows/docker-ci.yml
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 647 bytes | 647.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ayushi2323/Devops-lab
 ! [remote rejected] main -> main (refusing to allow a Personal Access
   Token to create or update workflow '.github/workflows/docker-ci.yml'
   without 'workflow' scope)
error: failed to push some refs to 'https://github.com/ayushi2323/Devops-lab'
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://[PASSWORD-HIDDEN]':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 647 bytes | 129.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ayushi2323/Devops-lab
 db2aaf6..9e7f00c main -> main
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ |
```



```
name: Docker CI/CD
```

```
on:|
```

```
  push:
```

```
    branches:
```

```
      - main
```

```
  pull_request:
```

```
    branches:
```

```
      - main
```

```
jobs:
```

```
  build-and-test:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout source code
```

```
        uses: actions/checkout@v4
```

```
      - name: Build Docker image
```

```
        run: docker build -t my-docker-app .
```

```
      - name: Run Docker container
```

```
        run: |
```

```
          docker run -d -p 5000:5000 --name my-container my-docker-app  
          sleep 5
```

```
      - name: Test application
```

```
        run: curl http://localhost:5000
```

```
      - name: Stop container
```

```
        run: docker stop my-container
```